



Seeed XIAO ESP32-C6, Wi-Fi 6, BLE 5.0, Zigbee, Thread, 512KB SRAM, 4MB Flash, UART, SPI, RISC-V

The Seeed Studio XIAO ESP32C6, based on ESP32-C6, offers advanced connectivity for Matter-compliant smart home projects. With 2.4 GHz Wi-Fi 6, BLE 5.0, Zigbee and Thread, it is ideal for safety-critical applications, supporting a wide range of smart h...

Product number SE-113991254

Weight 0.006kg



Product description

Seeed Studio XIAO ESP32C6

This cost-effective MCU is based on Espressif's ESP32-C6 and is characterised by its support for various wireless connectivities (2.4 GHz Wi-Fi 6, BLE 5.0, Zigbee and Thread), making it ideal for Matter-compatible smart home applications. With the thumb-sized design and single-sided mounting of the XIAO series, it is perfect for space-constrained projects. On-chip encryption ensures a desired level of security and enables a smooth development experience and fast time-to-market through easy integration with major IoT cloud platforms.

Features at a glance

- Enhanced connectivity: Combines 2.4 GHz Wi-Fi 6 (802.11ax), Bluetooth 5 (LE) and IEEE 802.15.4 radio connectivity, enabling the use of Thread and Zigbee protocols.
- Matter Native: Supports the development of Matter-compliant smart home projects thanks to its improved connectivity and achieves interoperability.
- Security Encrypted on Chip: Powered by ESP32-C6, it offers enhanced security through secure boot, encryption and Trusted Execution Environment (TEE).
- Outstanding RF performance: Features a built-in antenna with up to 80 m BLE/Wi-Fi range and reserves an interface for an external UFL antenna.
- Power consumption management: Features four operating modes, with the lowest at 15 μ A in deep sleep mode, and also supports lithium battery charging management.
- Classic XIAO design: Retains the classic XIAO design with a thumb-sized 21 x 17.5 mm form factor and single-sided mounting, ideal for space-constrained projects such as wearables.
- Two 32-bit RISC-V processors, with the high performance processor running at up to 160 MHz and the low power processor running at up to 20 MHz.
- 512KB SRAM and 4MB Flash on-chip for more programming memory and advanced IoT control capabilities.
- Supports UART, IIC and SPI serial communication ports.
- 15 total GPIO pins, including 11 digital I/Os for PWM pins and 4 analogue I/Os for ADC pins.

Technical specifications



- Processor: Espressif ESP32-C6 SoC, two 32-bit RISC-V processors, with a high performance processor up to 160 MHz and a low power processor up to 20 MHz.
- Wireless connectivity: Full 2.4 GHz Wi-Fi 6 subsystem, Bluetooth 5.0/ Bluetooth mesh, Zigbee, Thread, IEEE 802.15.4.
- On-chip memory: 512KB SRAM & 4MB Flash.
- Interfaces: 1x UART, 1x LP_UART, 1x IIC, 1x LP_IIC, 1x SPI, 11x GPIO (PWM), 7x ADC, 1x SDIO, 1x reset button, 1x boot button.
- Dimensions: 21 x 17.5 mm.
- Power supply: Input voltage (Type-C): 5V, input voltage (BAT): 4.2V, operating voltage: USB: 5V@9mA, BAT: 3.8V@9mA, battery charging current: 350mA.
- Energy consumption model (supply voltage: 3.8V): Modem sleep model: ~ 30mA, Light sleep model: ~ 2.5mA, Deep sleep model: ~ 15 ?A.
- Operating temperature: -40°C to 85°C.
- Size: 21 x 17.5 mm
- Input voltage (Type-C): 5V, input voltage (BAT): 4.2V
- Operating temperature: -40°C to 85°C

Scope of delivery

- 1x Seeed Studio XIAO ESP32C6

Product properties

Zolltarifnummer	84714900
Gross Weight (kg)	0.006
Country of Origin	China
Brand	seeed
Product ID	SE-113991254
Manufacturer ID	113991254



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